

Panini Linguistics Olympiad 2023

Round 1 (Senior), 5th March 2023
Question Booklet

Maximum marks: 100

Total time: 4 hours

- This question booklet has 10 pages, including this page. When the exam begins, please check and inform the invigilator immediately if your booklet has less than 10 pages.
- This question booklet has 5 problems. Each problem can have multiple assignments; the marks for each assignment are indicated next to the assignment, and the total marks for the problem are indicated next to the problem title.
- **It is not expected that you will solve every problem.** It is recommended to read all the problems before starting to solve any. You don't need to solve them in order.
- At the end of some problems, we've given you some extra information and interesting facts related to the problems in footnotes. They are not required to solve the problems but may sometimes be helpful.
- All problems will ask for short objective answers (e.g. match the following, multiple choice, one-line answers, etc.). Some problems may also ask for detailed explanations of your solutions. Your answer booklet has space for both.
 - Please write all your answers in the **specific space provided** in the answer booklet.
 - Space has also been provided for answers that require explanations. If you need extra space, you may continue outside the given space, or on the back of the sheet, or on a blank sheet that the invigilator will give you. You don't need to repeat the other answers in your explanations but you can refer to them by their number if you want to.
 - You don't need to copy the questions, problem statements, or any of the data in your answers, but you can refer to them in your own way as long as it is clear.
 - Please write the **problem number**, your **roll number**, and your **name** on **every sheet** of the answer booklet, including any extra sheets.
- You are not allowed to use any calculators, mobile phones, other electronic devices, books, notes, or other external sources of information during the exam. If you have any questions at any time, you should ask the invigilator, but you are not allowed to communicate with anyone else during the exam.

All the best!

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Problem 1: Nanti

(20 marks)

Here are some constructions in Nanti.¹

igemarite	his tapir
nobonkiti	my foot
iserine	his tobacco
oshiropitonkite	her needle
nogusine	my piping guan
pirirana	your blood
nobarotote	my raft
pibirane	your flashlight
pigotsirote	your knife
igabeharite	his bird-trap
ogito	her head
ibanko	his house
ogobitite	her pot
nogomaginarote	my woolly monkey
ogetane	her walnut
pitsitsihenkate	your soot

Assignment 1: (6 marks)

Translate from Nanti:

- | | |
|-------------|----------------|
| (a) kemari | (d) pibanko |
| (b) irirana | (e) ogotsirote |
| (c) ponkiti | (f) noserine |

Given below are some additional Nanti nouns.²

kuri	peach palm
pori	leg
saburi	machete

¹The Nanti language is spoken by ~250 people in Peru. The tapir is a large, pig-like animal. The piping guan is a South American bird species.

²The peach palm is a species of palm tree native to Central and South America.

Assignment 2: (6 marks)

Translate to Nanti:

- | | |
|-------------------|---------------------|
| (a) your leg | (d) raft |
| (b) my peach-palm | (e) woolly monkey |
| (c) his machete | (f) her piping guan |

Assignment 3: (1 marks)

In recent years, the word forms for the Nanti word kipatsi “land” changed after people began to buy and sell land. Based on the rules you have inferred, give the old and new forms for “your land”.

Assignment 4: (7 marks)

Explain your solution.

– Antara Raaghavi Bhattacharya

Problem 2: Romanian

(22 marks)

Here are some phrases in Romanian³ and their English translations:⁴⁵

copilul micuț al cetățeanului viteaz	the small child _M of the brave citizen _M
aptul cetățean al nepoatei moarte	the skilled citizen _M of the dead niece
camarazii curioși ai sătulelor cetățene	the curious comrades _M of the satiated citizens _F
vicleonii nepoți ai înalților copii	the cunning nephews of the tall children _M
camarada smeadă a mireșelor flămânde	the pale comrade _F of the hungry brides
nepoata vicleană a curiosului urs	the cunning niece of the curious bear _M
frumosul mânz al urșilor apți	the beautiful foal _M of the skilled bears _M
goalele nomade ale micuței camarade	the underdressed nomads _F of the small comrade _F
profesorul viteaz al hoților smezi	the brave teacher _M of the pale thieves _M
sătulii studenți ai hoatei înalte	the satiated students _M of the tall thief _F
mireasa înaltă a flămânzilor profesori	the tall bride of the hungry teachers _M
moartele verișoare ale frumoaselor profesoare	the dead cousins _F of the beautiful teachers _F

Assignment 1: (3 marks)

Translate to English:

³Romanian belongs to the Romance branch of the Indo-European family. It is spoken by approx. 24 million people primarily in Romania and Moldova.

⁴ș = sh in shop; ț = ts in cats; j = s in vision; ă = a in about; â and î are vowels.

_M = masculine; _F = feminine.

⁵A foal is a baby horse.

- (a) hoțul flămând al copilelor viteze
- (b) micuța cetățeană a mânjilor frumoși
- (c) studentele viclene ale înaltului nepot

Assignment 2: (6 marks)

Translate to Romanian. If multiple translations are possible, write only one:

- (a) the brave cousins_M of the dead foals_M
- (b) the satiated thief_F of the underdressed nomads_M
- (c) the curious cousin_F of the skilled child_F

Assignment 3: (4 marks)

Translate into Romanian in all possible ways:

- (a) the small citizens_M of the underdressed nomad_M

Assignment 4: (9 marks)

Explain how these phrases are structured in Romanian.

– Vlad A. Neasçu

Problem 3: Icari Dargwa

(13 marks)

Here are some sentences in Icari Dargwa⁶ and their English translations in random order:⁷

- | | |
|------------------------|---|
| A. dul mazi ucibda | i. I caught the cat. |
| B. dul nusa ucibdi | ii. I caught you _{pl} . |
| C. nusal du ucibdi | iii. The chief caught me. |
| D. nusal tati ucibda | iv. The chief caught the cat. |
| E. nusal tati ucibdi | v. The father caught the cat. |
| F. tatil mazi ucib | vi. The father caught us. |
| G. tatil nusa ucibda | vii. The girl saw the horse. |
| H. ul kurta ucibdi | viii. We caught the father. |
| I. ul nusa ucibdi | ix. You _{pl} caught me. |
| J. ustal du ucibda | x. You _{pl} caught the father. |
| K. ustal mazi ucib | xi. You _{sg} caught the fox. |
| L. rirsil jabu čidažib | xii. You _{sg} caught us. |

Assignment 1: (6 marks)

Determine the correct correspondences.

Assignment 2: (2 marks)

Translate into English:

- (a) rirsil nusa ucibdi
- (b) mazil du čidažibda

Assignment 3: (3 marks)

Translate into Icari Dargwa:

- (a) The father saw the girl.
- (b) The horse caught me.
- (c) We saw you_{sg}.

Assignment 4: (2 marks)

Explain your solution.

– Kazune Sato

⁶Icari Dargwa is a variant of the Dargwa language, which belongs to the Northeast Caucasian (Nakh-Dagestani) family. It is spoken by approx. 497 thousand people in the Republic of Dagestan, Russia.

⁷c = ts in tsar, č = ch in chop, ž = s in vision.

Problem 4: Epun

(25 marks)

Here are some sentences in Epun⁸ and their English translations:⁹

Midebiniz sminus fa vakvepiz.
Chavetius f imnikiniz gub midep.
Tak huspat asa smin?
Asa hahielogus chegodiniz.
Guza hakakolgu gub zenadin.
Chupanibu fa pelikin.
Osozeu uza imnipiz.
Tak hagupat gub imnidin kakol?
Afa bozudin cherehelgu gub vletop.
Tak chupat f imnikin osoze?
Nisa osozenis fa chegodopiz.
Af imnipiz bozudiniz havetigus.
Zaddiliniz hapanibus.
Makohin binapu.

Some kings kiss the walruses_M.
The cats_M won't love this queen.
What have they_M kissed?
Some dogs_F haven't looked at them.
This girl hasn't seen her.
The turtle_M won't come.
He doesn't catch some cats_F.
What has this cat_F seen?
The teacher_F will write this letter.
What will the cat_M catch?
We catch the dogs_M.
Some teachers_F have loved the cats_F.
Some men have come.
The boy returns.

Assignment 1: (3 marks)

Translate into English:

- (a) Tak hugupat panib?
- (b) Uza havetigu guza.
- (c) Pelikop nisa hakakolnis.

Assignment 2: (8 marks)

Translate into Epun:

- (a) The walrus_F will kiss us.
- (b) This man hasn't written a letter.
- (c) Some turtles_F don't look at a woman.
- (d) The boys have loved the queens.

Assignment 3: (6 marks)

Translate into Epun in all possible ways:

- (a) What do they love?

⁸Epun is an invented language, deliberately devised to contain constructions which violated universal grammatical principles.

⁹ch = ch in chop. The subscripts M and F refer to masculine and feminine, respectively.

(b) What will return?

Assignment 4: (8 marks)

Explain your solution.

– Vlad A. Neacșu (inspired by a problem created by Adina Dragomirescu)

Problem 5: Old Persian

(20 marks)

Old Persian was spoken from 600-300 BCE in Achaemenid Persia. It was written in cuneiform¹⁰. Since the medieval times travelers had noticed the Old Persian inscriptions in cuneiform carved in the palaces and ruins of Persepolis, a city in Iran. However, the script remained undeciphered till 18th century. In 1802, Georg Friedrich Grotefend, a German philologist, managed to (almost correctly) decipher most of the script starting from two simple inscriptions, known as the Niebuhr inscriptions 1 and 2.¹¹

Niebuhr inscription 1:

[illegible]

Niebuhr Inscription 2:

«II Ṛ K m E Ṛ m \ «II Ṛ m K K I Ṛ K \
 ቱ ቱ E Ṛ Ṛ \ «II Ṛ m K K I Ṛ K \ «II Ṛ m K
 K I Ṛ K Ṛ m Ṛ \ Ṛ m E K ቱ Ḡ Ṛ \ «II Ṛ m
 K K I Ṛ K \ Ḡ Ḡ Ṛ \ Ḡ «II m Ṛ Ṛ Ṛ Ṛ Ṛ K

Based on the repetition patterns in the two inscriptions and context in which they occur, Grotefend correctly guessed that the translation of the first inscription should be:

¹⁰Cuneiform was a writing system produced using wedge-shaped marks in clay tablets, inherited from the Sumerian Empire.

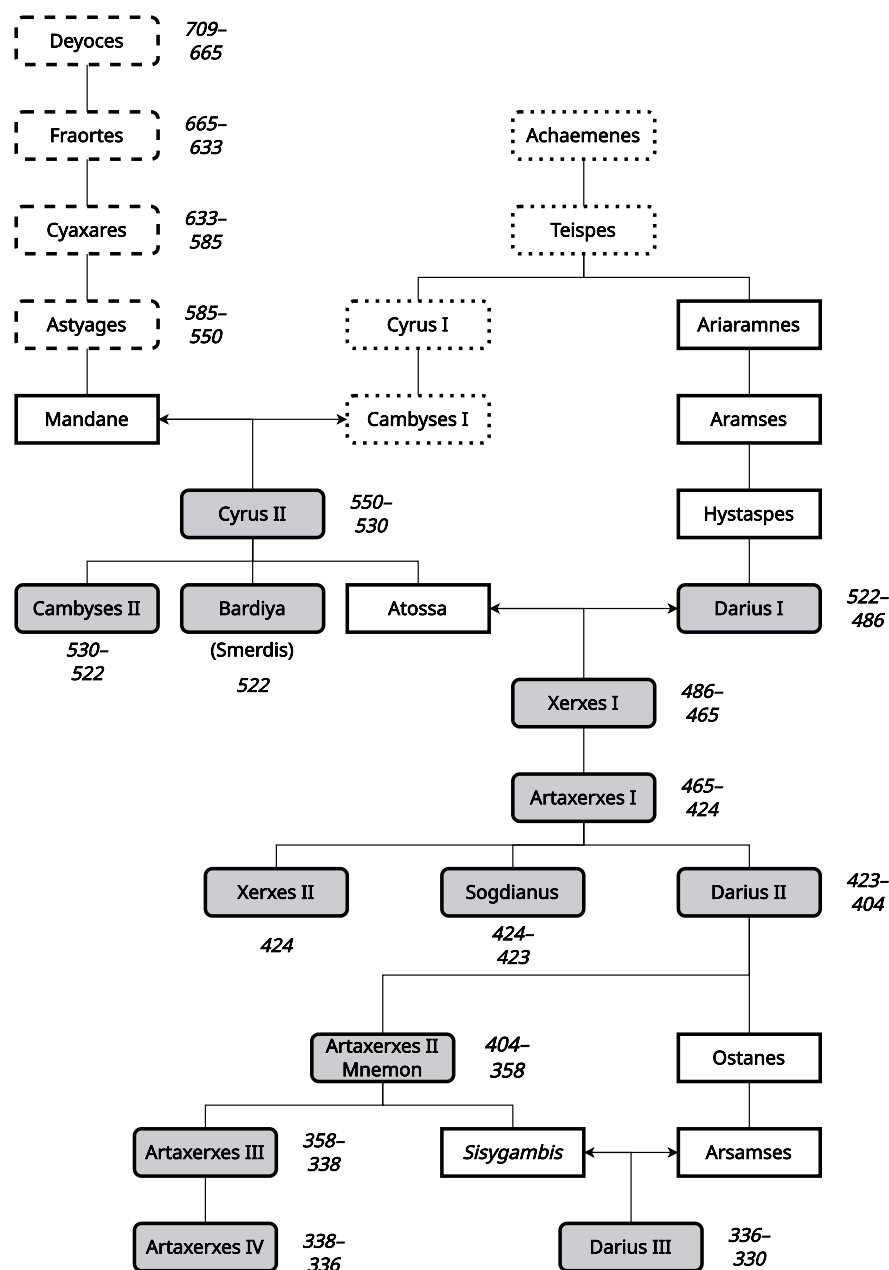
¹¹ Slightly simplified for this problem.

Translation 1: *A, the Great King, King of Kings, King of countries, son of B, an Achaemenian, who built this Palace.*

Note: *Achaemenian* refers to a descendant of Achaemenes.

Grotefend deciphered the above words without deciphering the letters. He could see that the second inscription also mentions two names, let's say, C and D (in that order).

At first, Grotefend was unable to decipher the names - A, B, C and D. However, the family tree of the Achaemenid dynasty gave him enough clues to pin down the names. The tree is shown below. The names follow the modern English spellings that closely reproduce the Greek pronunciations of the Old Persian words. The Achaemenid kings (descendants of Achaemenes who also reigned) are shown in shaded boxes. The reigning periods of all the kings have been shown as well.



Assignment 1: (5 marks)

Translate Niebuhr inscription 2 into English (you can use C and D for the names).

Assignment 2: (7 marks)

Identify A, B, C, and D. Explain how you arrived at your answer.

Assignment 3: (5 marks)

Give the Old Persian cuneiforms for the following English words:

- (a) King
- (b) Son
- (c) Achaemenian

Assignment 4: (3 marks)

What do the following cuneiform symbols stand for:

- (a) 
- (b) 
- (c) 

– Monojit Choudhury

This is the end.